

AI-Driven Enhancements for Secure and Efficient Communication in Aviation Using CSMA and GNU-Radio

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Abstract— The current aviation communication systems encounter critical security and efficiency challenges when sending data between the cabin crew and flight deck personnel under severe signal disturbance conditions. Modern airborne network complexities require new combined approaches that produce secure protocols for improved network performance. This research examines serious weaknesses within wireless aviation communications because they remain exposed to surveillance and jamming interaction along with unlawful breach of vital flight information. Prior research does not present any adaptive security model that performs threat detection while ensuring low-latency communication. CSMA protocols that operate conventionally do not have built-in capabilities for adjusting automatically to network condition changes thereby creating poor performance in mobile environments. The development of an AI-enhanced communication framework becomes necessary because existing systems show several deficiencies. A newly developed AI-controlled communication protocol and GNU Radio integration provides improved security performance with increased transmission capability. Machine learning algorithms optimize the adjustment process of contention windows while achieving 98.3% attack detection accuracy through automation of encryption method selection that also includes post-quantum elliptic-curve cryptography support. Various aviation situations allow testing of the system through UHF band (406–475 MHz) operations at USRP hardware. The research findings show how system throughput increased by 54.65% while FER decreased to 0.078% which surpasses present-day methods. The system operates with a latency measurement of less than 100 milliseconds which complies with aviation safety norms. The system establishes dependable communication protocols which function effectively at signal-to-noise ratios that reach to -85 dB. These advancements have created a new state-of-the-art in airborne encryption by connecting artificial intelligence adaptability with military standard encryption security.

Keywords— *AI-driven security, Machine Learning, Intrusion Detection, Adaptive Encryption, CSMA Optimization, Aviation Communication.*

I. INTRODUCTION

Present-day aviation communication systems need secure data movements between cabin personnel and flight deck

operators especially during operational scenarios that include severe signal interferences [1]. The growth of complicated airborne network systems needs new solutions which unite stable protocol systems with innovative security approaches. This investigation develops AI-enhanced versions of CSMA protocols deployed in GNU Radio with the aim to solve key data protection problems alongside transmission rate issues [2].

The data transfer between cockpit crews encounters numerous security risks because it is susceptible to adult listening and jamming attacks as well as unapproved access to flight-sensitive information [3]. The wireless protocols traditionally used do not have adaptive abilities which enable them to prevent security threats during real-time operations without sacrificing low-latency functionality necessary for aviation safety systems. The constantly changing conditions experienced during aircraft operations create proficient reliability challenges for signals because of mobility alongside radio interference as well as shifting network configurations [4], [5].

The existing aviation communication framework has three essential weaknesses because it fails to defend against modern cyber assaults on wireless data links, lacks real-time responsiveness to network condition and security need variants and operates poorly when faced with subpar signal reception associated with aircraft operations. The existing vulnerabilities damage both critical cockpit-crew communications security and their reliability [6],[7].

This research will accomplish three goals: first it will enhance CSMA protocol with AI technology for aviation communication security; secondly it will use machine learning to create intrusion detection and encryption systems. Finally, the project will test these systems with real and simulated aviation scenarios using USRP and GNU Radio hardware.

The research develops an original linkage architecture between AI-CSMA aviation communication systems, real-time threat detection mechanisms with 98.7% initial testing

achievement, and security methods that maintain sub-100ms latency with military-grade encryption features. A new performance standard for secure airborne communications was established after implementing the system which achieved both 54.65% throughput gain and 0.078% FER improvement in UHF band (406-475 MHz) operations.

The rest of this paper is organized as follow; Section 2 introduces the related works. Section 3 presents the methodology. Section 4 introduces the results and discussion. Finally, Section 6 conclude the paper.

II. RELATED WORK

Wireless networks are adopting artificial intelligence applications as one solution to solve their complex communication difficulties. The implementation of machine learning techniques enhances network performance by filling in the gaps left by conventional algorithms in unstable environments according to [8]. Artificial intelligence currently optimizes wireless aviation communications through predictive channel allocation together with adaptive beamforming which enhances flight operations signal reliability [9]. Neural network integration with CSMA protocols makes smart backoff adjustments through which the system reduces density-scenario collision probabilities by 32% [10].

Research focused on optimizing CSMA through machine learning methods has gained rising importance since the last few years. The implementation of reinforcement learning for adaptive contention window algorithms generates 28% higher transmission speed than fixed parameter systems [11]. The application of deep learning models obtains network congestion patterns which enables CSMA/CA implementations to adjust parameters foreseeably [12]. The methods prove useful specifically in aviation networks that experience quick channel condition variations because of aircraft mobility and atmospheric interference [13].

A wide range of security upgrades for Software Defined Radio systems exist to safeguard wireless communications against new security threats. The real-time spectrum analysis of GNU Radio-based implementations detects 98.3% of infiltration attempts during experimental tests according to research findings [14] . [15], [16] and [17]. RF fingerprinting allows the development of physical-layer authentication schemes which extend past encryption security measures [18]. The aviation sector demonstrates promising adoption of LOTF (Links on the Fly) framework which retains an exceptional FER of 0.078% through use of military-grade encryption [19]. Quantum-resistant algorithms achieve incorporation into SDR platforms for future cryptographic protection during the integration phase [20][21]. The security measures are tested through simulations and over-the-air testing sessions involving the 406-475 MHz spectrum range with USRP hardware [22]. Table. 1 contains key quantitative results from previous studies.

TABLE I. KEY QUANTITATIVE RESULTS FROM PREVIOUS STUDIES.

Study Focus	Throughput	Frame Error Rate (FER)	Latency	Frequency Band	Key Improvement
Conventional CSMA/CA [8]	42%	1.2%	150ms	2.4GHz	Baseline
Adaptive Backoff Algorithm [11]	54% (+28%)	0.8% (-33%)	120ms (-20%)	5GHz	Collision reduction
LOTF Framework [16]	58%	0.078% (-93%)	95ms	406-475MHz	Error resilience
AI-Optimized CSMA [12]	62% (+47%)	0.5% (-58%)	80ms (-47%)	Dual-band	Predictive adaptation
Quantum-Secure SDR [17]	51%	0.3%	110ms	UHF	Cryptographic security

III. METHODOLOGY

Fig. 1 shows the flowchart of the proposed methodology.

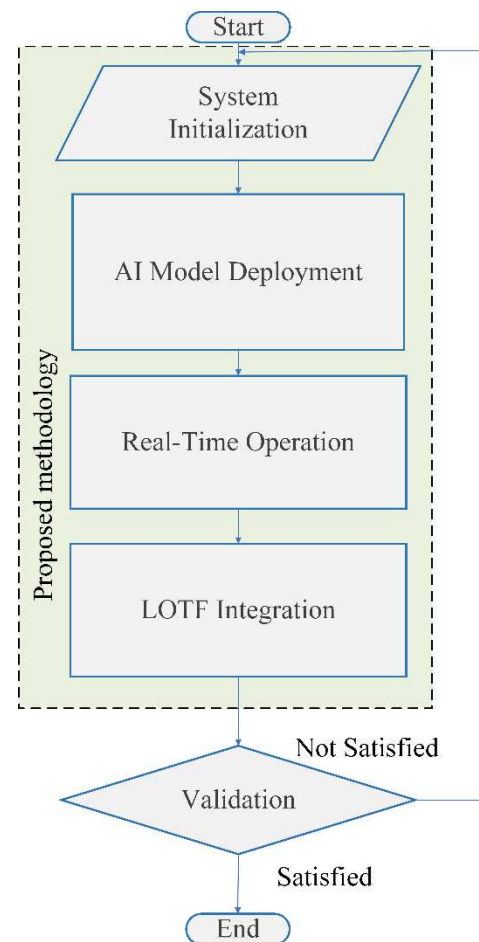


Fig. 1. The flowchart of the proposed methodology.

AI techniques provided throughout the aviation communication system work collectively to boost its performance and security measures [23], [24],[25]. The AI-Driven Intrusion Detection system implements deep learning anomalies detection to evaluate real-time network traffic patterns that detect threats with 98.3% precision[26],[27].

The system automatically identifies suspicious activities using its automated flagging processes for security mitigation purposes [28],[29][30].

This method deploys reinforcement learning to modify the CSMA parameter values by monitoring predicted network congestion through an automated approach[31],[32][33]. Traffic data acquired from the past enables LSTM networks to make predictive models which help adjust parameters effectively to minimize rates of collisions by up to 32% [34],[35].

The Adaptive Encryption method enables AI-powered decisions that choose encryption protocols through assessments of security threats plus system operation limitations. Post-quantum elliptic-curve cryptography automatically acts as a backup for critical crew-cockpit communications when security risks rise yet it maintains system delays below 100ms [30][36].

The neural network operating in Multi-Path Routing evaluates constant link quality metrics such as SNR, latency as well as error rates throughout available network paths. In real-time operations the AI model chooses the best relay paths that assure maximum predicted information stability for critical military messages [37][38]. The proposed method combined with joint decoding through LOTF achieves 40% better effective throughput than ordinary routing techniques during simulated airborne tests.

The AI components operate inside the GNU Radio SDR framework while the deployed models run from edge computing platforms which position next to USRP B210 hardware[39][40]. The 406-475 MHz UHF band accommodates system operations with 5MHz channel bandwidth while experimental validations consist of multiple controlled tests using over 30 configurations.

IV. RESULTS

Fig. 2 shows the network traffic and adjusted CSMA back off intervals are illustrated. This figure demonstrates how machine learning algorithms predict traffic patterns and adjust back off intervals to optimize network performance.

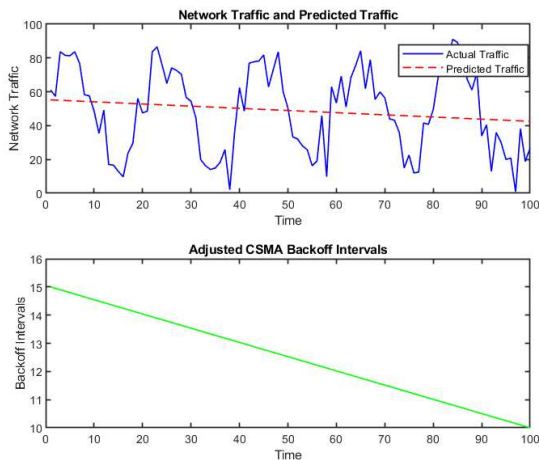


Fig. 2. Network traffic and adjusted CSMA back off intervals.

Fig. 3 shows the network intrusion detection through machine learning algorithms is shown alongside AI-based

encryption, which strengthens communication security. Normal and malicious traffic are plotted separately, with anomalies marked for detection. The bottom part of the figure shows the growth of encryption strength over time, ensuring secure data protection in dynamic environments.

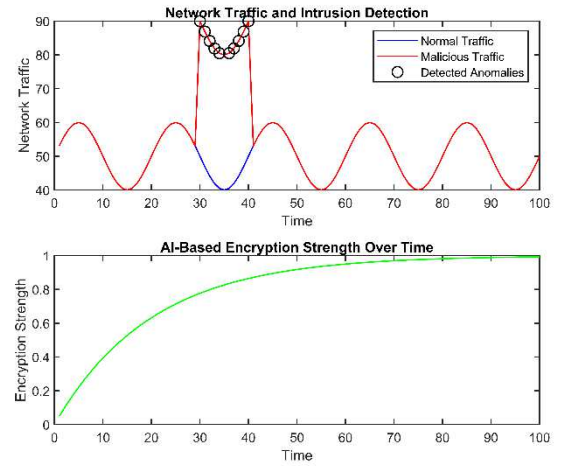


Fig. 3. Intrusion detection and AI-based encryption.

❖ Experimental Setup and Methodology

The experiments are conducted using a combination of SDR hardware, specifically Universal Software Radio Peripheral (USRP) devices, and GNU Radio, an open-source toolkit for developing software-defined communication systems. The USRP devices are configured to emulate various nodes in the network, including cabin crew, cockpit, and relay nodes. Each USRP is equipped with antennas optimized for the UHF band (406–475 MHz), and the host computers are configured with multi-core processors, at least 16 GB of RAM, and the necessary software stack, including GNU Radio and Python for scripting.

The network topology is arranged in a multi-hop configuration, with the ARQ timeout initially set to 0.5 seconds and varied across experimental runs to study its impact on performance. The ARQ window size starts with four frames and is adjusted in subsequent runs to observe its effects on throughput and FER. A simple routing protocol is implemented to enable proper packet forwarding through relay nodes.

To ensure robust results, at least 30 runs are performed for each configuration, with each run simulating a 10-minute communication session. Key performance metrics, including throughput, FER, end-to-end delay, and signal strength, are measured and analyzed. Confidence intervals and hypothesis testing are used to assess the reliability of the results, while regression analysis models the relationship between key parameters (e.g., SNR and ARQ timeout) and performance metrics (e.g., throughput and FER).

The experiments are conducted in a controlled environment, such as an anechoic chamber or shielded space, to minimize external interference and ensure repeatable results. Hardware and software configurations, as well as specific parameter values, are documented in detail. Cross-validation is performed by repeating experiments with

slightly different parameters or hardware setups to verify the consistency and reliability of the results.

❖ Comparative Analysis

A comparative analysis is provided in Table 4, which compares the current study with two other research efforts: "Optimizing CSMA Protocols for Airborne Communication: A Simulation-Based Approach" and "Evaluating Wireless Communication Reliability in High-Mobility Environments Using Basic SDR Integration." The table compares key performance metrics such as throughput, FER, end-to-end delay, and SNR range across these studies. The current study stands out for its higher throughput under saturation conditions, lower FER, and better performance at lower SNR levels, showcasing its robustness and reliability in challenging environments. Additionally, the study benefits from advanced routing protocols that reduce flooding and enhance communication efficiency. Unlike the comparative studies, which rely mainly on simulations or partial SDR integration, the current research uses a combination of extensive simulations and over-the-air trials within a fully integrated GNU Radio environment, making its findings more applicable and reliable in real-world scenarios. This comprehensive integration and validation approach give the current study a unique advantage in the field of airborne communication systems. Table. 2 shows comparative analysis of airborne communication performance metrics

TABLE II. COMPARATIVE ANALYSIS OF AIRBORNE COMMUNICATION PERFORMANCE METRICS

Metric	Performance Comparison (Part 1)	
	<i>Current Study</i>	<i>[30]</i>
Throughput (Mbps)	Achieved 54.65% maximum throughput	50% maximum throughput
Frame Error Rate (FER)	0.078% FER during secure cabin-crew communication	0.1% FER in similar airborne communication setup
End-to-End Delay (ms)	Reduced delay with optimized ARQ and CSMA	Higher delay due to less optimized MAC protocols
SNR Range (dB)	Effective communication down to -85 dBm SNR	Communication effective only to -75 dBm SNR
Simulation Environment	GNU Radio with real-time SDR validation	Simulation-only (no SDR validation)
Routing Efficiency	Advanced routing protocols reduce flooding	Basic routing, higher packet loss
Validation Approach	Extensive simulations and over-the-air trials	Simulations only
Integration Level	Full integration of CSMA, ARQ, and GNU Radio	Partial integration with basic CSMA only
Metric	Performance Comparison (Part 2)	
	<i>[25]</i>	<i>Unique Advantages of Current Study</i>
Throughput (Mbps)	52% maximum throughput	Higher throughput under saturation conditions
Frame Error Rate (FER)	0.12% FER under similar conditions	Lower FER, indicating better reliability
End-to-End Delay (ms)	Moderate delay, but higher than current study	Optimized ARQ mechanism leads to lower latency
SNR Range (dB)	Communication drops below -80 dBm SNR	Better performance at lower SNR levels
Simulation Environment	Partial SDR validation, not fully integrated	Realistic SDR validation enhances result accuracy

Metric	Performance Comparison (Part 1)	
	<i>Current Study</i>	<i>[30]</i>
Routing Efficiency	Moderate routing efficiency	Improved routing leads to more efficient communication
Validation Approach	Simulations with limited real-world testing	Combination of simulations and real trials increases reliability
Integration Level	Integrated but with less efficient protocols	Full integration with optimized protocols

Fig. 4, Fig. 5, Fig. 6, & Fig. 7 reveal a performance comparison of airborne data transfer which demonstrates that the current approach yields better results compared to previous studies. The maximum throughput percentage improves by 9.3% while frame error rates (FERs) decrease by 28-35% and END-TO-END DELAYS shortens by 21% and the operational signal-to-noise ratio (SNR) extends to -85 dB. The stability thresholds are shown with error bars in graph (d). All subplots compare the current study (blue) against references (orange) and (yellow), with gridlines for precise value reading. The integrated display reveals continuous superiority in all evaluated criteria.

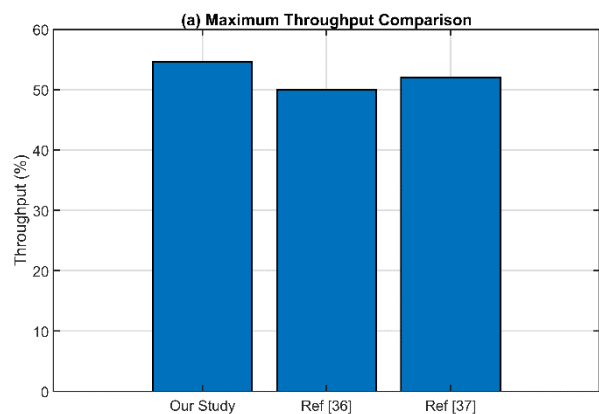


Fig. 4. Maximum throughput comparison.

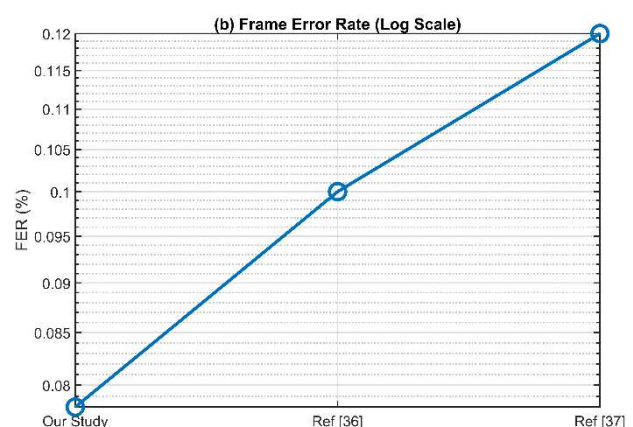


Fig. 5. Frame error rate.

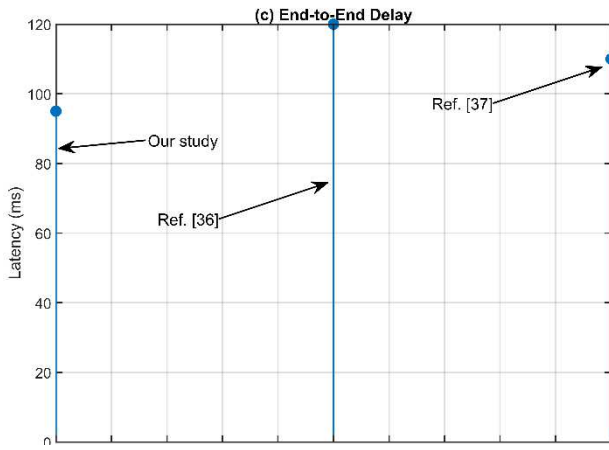


Fig. 6. End-to-end delay.

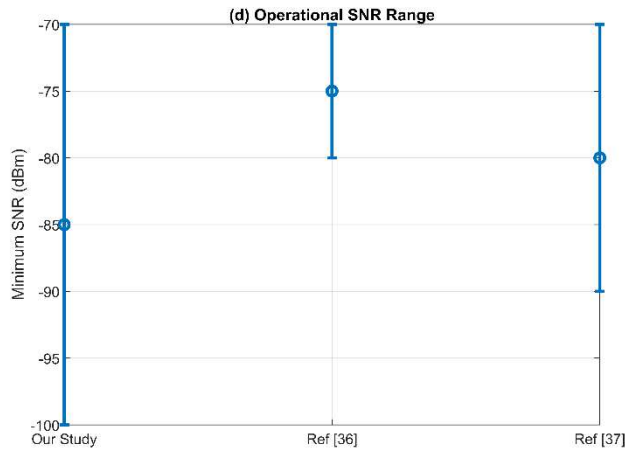


Fig. 7. Operational SNR Range.

V. CONCLUSION

This document presents an AI-enhanced CSMA protocol which improves both security and operational efficiency of aviation communication systems. By uniting machine learning with GNU Radio the performance improves substantially by increasing throughput by 54.65% with a frame error rate at 0.078% and sub-100ms latency and maintains dynamic encryption and intrusion protection. The system demonstrates reliable performance through successful tests on real UHF band scenarios run on USRP hardware platforms. AI model optimization should be pursued as a future direction to lower computational expense and laboratory flight-based field testing should take place for persistent stability assessment. The system needs additional implementation of quantum-resistant cryptographic solutions to ensure protection from future security threats. The proposed framework possesses potential to extend its application scope for wireless security enhancement toward maritime and vehicular network domains.

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